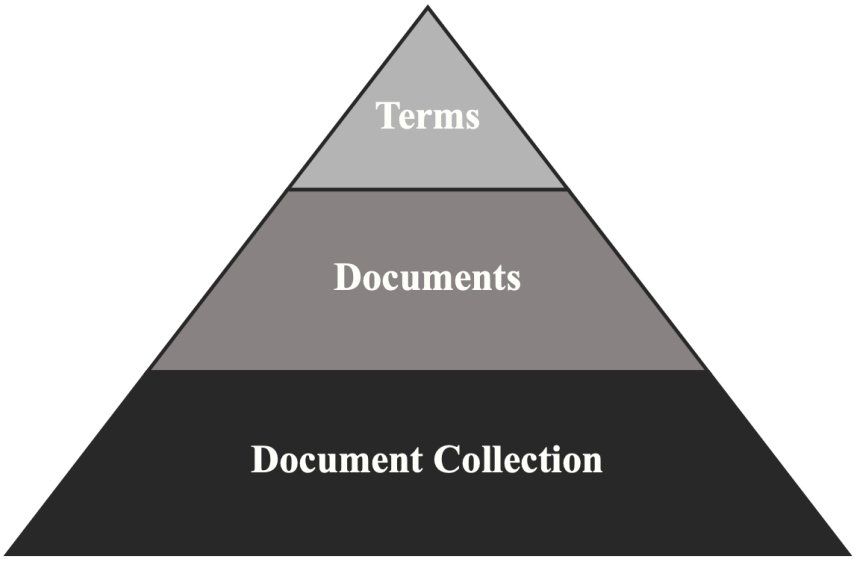
Text Processing

CSEC 6 - Natural Language Processing



Hierarchy of terms and documents

corpus

A collection of documents.

DOCUMENT 1
My favorite dog is fluffy and tan.

DOCUMENT 2
the dog is brown and cat is brown

DOCUMENT 3
My favorite hat is brown and coat is pink

DOCUMENT 4
My dog has a hat and leash. ♥

DOCUMENT 5
He has a fluffy coat and brown coats.

DOCUMENT 6
The dog is brown and fluffy & has a brown coat.

DOCUMENT 7
MY dog is white with brown spots.

DOCUMENT 8
The white dog has a Pink coat and the Brown dog is fluffy

DOCUMENT 9
The 3 fluffy dogs AND 2 brown hats are my favorites!

DOCUMENT 10
MY fluffy dog has a white coat and hat .

document

is typically made up of many characters

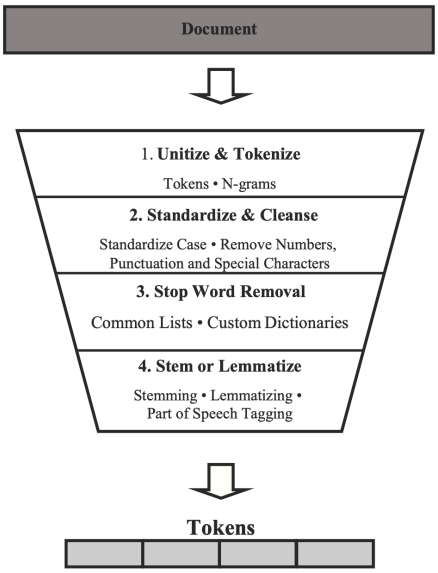
characters

are combined to form words or terms in a given language

vocabulary

collection of terms

Preprocessing Process



The text data pre-processing process

Unitize

The first step involves the choice of the ***unit of text*** to analyze and the separation of the text based on the unit of analysis. This unit could be a word; however, in other cases, it may be a grouping of words or a phrase. ***Single words*** are the simplest choice and make a good starting point.

Tokenization

The process of separating text into a more usable form known as tokens, which are single words or groups of words that serve as the features for analysis.

DOCUMENT 1
[My] [favorite] [dog] [is] [fluffy] [and] [tan] [.]

DOCUMENT 2
[the] [dog] [is] [brown] [and] [cat] [is] [brown]

DOCUMENT 3
[My] [favorite] [hat] [is] [brown] [and] [coat] [is] [pink] [.]

DOCUMENT 4
[My] [dog] [has] [a] [hat] [and] [leash] [.] [♥]

DOCUMENT 5
[He] [has] [a] [fluffy] [coat] [and] [brown] [coats] [.]

DOCUMENT 6
[The] [dog] [is] [brown] [and] [fluffy] [&] [has] [a] [brown] [coat] [.]

DOCUMENT 7
[MY] [dog] [is] [white] [with] [brown] [spots] [.]

DOCUMENT 8
[The] [white] [dog] [has] [a] [Pink] [coat] [and] [the] [Brown] [dog] [is] [fluffy]

DOCUMENT 9
[The] [3] [fluffy] [dogs] [AND] [2] [brown] [hats] [are] [my] [favorites] [!]

DOCUMENT 10
[MY] [fluffy] [dog] [has] [a] [white] [coat] [and] [hat] [.]

n-grams

N-grams are an alternative to single words in the tokenization process. N-grams are tokens; they are consecutive word sequences with length n.

For instance, ***bigrams*** are tokens composed of **two side-by-side words**; a **single word** is known as a ***uni-gram***.

n-grams

The first token is:
My favorite dog is fluffy and tan.

The second token is:
My favorite dog is fluffy and tan.

The final token is:
My favorite dog is fluffy and tan.

The bigram representation of Document 1 is:
My favorite
favorite dog
dog is
is fluffy
fluffy and
and tan
tan

Standardization and Cleaning

Standardization and cleaning involve transforming text tokens into a consistent format to ensure accurate comparison across documents. This process typically begins by converting all text to lowercase to prevent identical words with different capitalization from being treated as unique terms. Additionally, it involves removing numbers, punctuation, special characters (such as ampersands or emojis), and excessive whitespace, resulting in a refined dataset where only the core textual content remains for analysis.

DOCUMENT 1
[my] [favorite] [dog] [is] [fluffy] [and] [tan]

DOCUMENT 2
[the] [dog] [is] [brown] [and] [cat] [is] [brown]

DOCUMENT 3
[my] [favorite] [hat] [is] [brown] [and] [coat] [is] [pink]

DOCUMENT 4
[my] [dog] [has] [a] [hat] [and] [leash]

DOCUMENT 5
[he] [has] [a] [fluffy] [coat] [and] [brown] [coats]

DOCUMENT 6
[the] [dog] [is] [brown] [and] [fluffy] [&] [has] [a] [brown] [coat]

DOCUMENT 7
[my] [dog] [is] [white] [with] [brown] [spots]

DOCUMENT 8
[the] [white] [dog] [has] [a] [pink] [coat] [and] [the] [brown] [dog] [is] [fluffy]

DOCUMENT 9
[the] [fluffy] [dogs] [and] [brown] [hats] [are] [my] [favorites]

DOCUMENT 10
[my] [fluffy] [dog] [has] [a] [white] [coat] [and] [hat]

Stop Word Removal

to drop frequently used filler words, or stop words, which add no value to the analysis.

Standard stop word dictionaries eliminate many of the most common words in a given language, but sometimes we also want to drop domain- or project-specific words or tokens.

DOCUMENT 1
[favorite] [dog] [fluffy] [tan]

DOCUMENT 2
[dog] [brown] [cat] [brown]

DOCUMENT 3
[favorite] [hat] [brown] [coat] [pink]

DOCUMENT 4
[dog] [hat] [leash]

DOCUMENT 5
[fluffy] [coat] [brown] [coats]

DOCUMENT 6
[dog] [brown] [fluffy] [brown] [coat]

DOCUMENT 7
[dog] [white] [brown] [spots]

DOCUMENT 8
[white] [dog] [pink] [coat] [brown] [dog] [fluffy]

DOCUMENT 9
[fluffy] [dogs] [brown] [hats] [favorites]

DOCUMENT 10
[fluffy] [dog] [white] [coat] [hat]

Stemming and Lemmatization

Stemming and lemmatization involve breaking words down to their root word. Both methods combine words that contain the same root into a single token to reduce the number of unique tokens within the analysis set.

Syntax

Syntax studies sentence structure and the rules governing how words combine to form meaningful expressions.

Semantics

Semantics focuses on the study of meaning in language. It explores how words, phrases, and sentences convey information and represent concepts.

Stemming

Stemming involves the removal of a word's suffix to reduce the size of the vocabulary.

DOCUMENT 1
[favorit] [dog] [fluffi] [tan]

DOCUMENT 2
[dog] [brown] [cat] [brown]

DOCUMENT 3
[favorit] [hat] [brown] [coat] [pink]

DOCUMENT 4
[dog] [hat] [leash]

DOCUMENT 5
[fluffi] [coat] [brown] [coat]

DOCUMENT 6
[dog] [brown] [fluffi] [brown] [coat]

DOCUMENT 7
[dog] [white] [brown] [spot]

DOCUMENT 8
[white] [dog] [pink] [coat] [brown] [dog] [fluffi]

DOCUMENT 9
[fluffi] [dog] [brown] [hat] [favorit]

DOCUMENT 10
[fluffi] [dog] [white] [coat] [hat]

Stemming

DOCUMENT 9 (BEFORE STEMMING)
[fluffy] [dogs] [brown] [hats] [favorites]

DOCUMENT 9 (AFTER STEMMING)
[fluffi] [dog] [brown] [hat] [favorit]

Porter’s algorithm

The most common English stemmer

Lemmatization

Similar to stemming, but it uses the word's part of speech to more accurately group words to their root or "lemma".

DOCUMENT 1
[favorite] [dog] [fluffy] [tan]

DOCUMENT 2
[dog] [brown] [cat] [brown]

DOCUMENT 3
[favorite] [hat] [brown] [coat] [pink]

DOCUMENT 4
[dog] [hat] [leash]

DOCUMENT 5
[fluffy] [coat] [brown] [coat]

DOCUMENT 6
[dog] [brown] [fluffy] [brown] [coat]

DOCUMENT 7
[dog] [white] [brown] [spot]

DOCUMENT 8
[white] [dog] [pink] [coat] [brown] [dog] [fluffy]

DOCUMENT 9
[fluffy] [dog] [brown] [hat] [favorite]

DOCUMENT 10
[fluffy] [dog] [white] [coat] [hat]

Lemmatization

DOCUMENT 9
[fluffy] [dogs] [brown] [hats] [favorites]

DOCUMENT 9 (AFTER STEMMING)
[fluffi] [dog] [brown] [hat] [favorit]

DOCUMENT 9 (AFTER LEMMATIZATION)
[fluffy] [dog] [brown] [hat] [favorite]

Lemmatization

DOCUMENT 9
[fluffy] [dogs] [brown] [hats] [favorites]

DOCUMENT 9 (AFTER STEMMING)
[fluffi] [dog] [brown] [hat] [favorit]

DOCUMENT 9 (AFTER LEMMATIZATION)
[fluffy] [dog] [brown] [hat] [favorite]

Word	Part of speech	Lemmatization
Fluffy	Adjective	Fluffy
Fluffier	Adjective	Fluffy
Fluffiest	Adjective	Fluffy
Fluffiness	Noun	Fluffiness
Fluffily	Adverb	Fluffily

Word	Part of speech	Lemmatization
Favorite	Noun/adjective	Favorite
Favorites	Noun	Favorite
Favorited	Verb	Favorited

Part-of-Speech (POS) Tagging

Part-of-speech tagging involves labeling tokens or words by their part of speech. Two of the most popular tag sets in English are the Brown Corpus and the Lancaster-Oslo-Bergen (LOB) Corpus. A newer tag set, the Penn Treebank, was developed in 1989 and has over 7 million words tagged by their parts of speech.

Thank you!

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Joseph Jessie S. Oñate, MSc.

Assistant Professor IV